

Automated Tooth Segmentation in X-ray Images using Attention Integrated U-Net++ Model

Julakanti Sai Yaswanth

Dept. of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India.
bl.en.u4cse21074@bl.students.amrita.edu

Kanderi Johith Kumar

Dept. of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India.
bl.en.u4cse21086@bl.students.amrita.edu

Rimjhim Padam Singh

Dept. of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India.
ps_rimjhim@blr.amrita.edu

Abstract—Oral functioning, the ability to perform various oral functions, pain-free and without being ill, is one of the indicators of overall well-being and life satisfaction. Even though these diseases are mostly preventable, a majority of individuals all over the world suffer from periodontal disease, dental caries, and oral malignancies to an estimated of 5 billion people worldwide. However, there are still use of traditional radiographic techniques like periapical and panoramic radiography that assist in diagnosis. A better diagnosis is enhanced through the use of modern techniques in imaging such as MRI, ultrasonography and CBCT imaging. Deep learning-based solutions enhance diagnostic capabilities through dental segmentation in images since they learn from unmodified data in contrast to traditional machine learning and deep learning techniques which produce inferior results when dealing with massive datasets. Therefore, in an effort to solve the problem of efficient dental segmentation, an encoder-decoder based tooth segmentation model based on U-Net++ architecture modified using Efficient Channel Attention Network is proposed that aims at enhancing the accuracy of dental diagnosis. The paper also focus of the performance of different state-of-art CNN techniques namely, U-Net, FCNs, ENet, U-Net++, U-Net 3+, SwiftNet on the standard Tufts Dental dataset for model selection and comparisons. The proposed model outperforms other segmentation models and achieves the best mean average precision of 96.4% and a dice coefficient of 90%.

Keywords - Convolutional Neural Networks, Deep Learning Tufts Dental database, Models MRI, U-Net++, Segmentation

I. INTRODUCTION

Nowadays, dental health tends to be a major indicator of overall health and life satisfaction. The capacity to articulate, smile, discern tastes, touch, chew, ingest, and employ gestures to convey a range of emotions while avoiding pain, discomfort, and craniofacial diseases is all included in oral health. The most frequent oral disorders are periodontal disease, often known as gum disease, dental caries, tooth decay, and oral cancers. Despite being mostly avoidable, oral disorders are persistent, impacting over 3.5 billion people globally. Dental caries, the most common disease in the world, is becoming more widespread. Radiographs are helpful tools that may be used in between a clinical examination to diagnose dental problems. Panoramic and periapical radiographs are often used and

provide information for routine treatments. However, limitations arise when projecting the entirety or a portion of the oral cavity onto a two-dimensional image plane. To address these issues and provide additional information needed for the diagnosis or treatment planning of specific patients, new three-dimensional imaging techniques are needed. There is the cone beam computed tomography, abbreviated as CBCT, the magnetic resonance imaging known as MRI, and ultrasound imaging. The entire orofacial region, encompassing the jaws, teeth, sinuses, and temporomandibular joint (TMJ) is visible on panoramic radiographs. These images are highly informative for illustrating abnormalities or phases of oral growth, as well as acting as a preliminary assessment for widespread illness or numerous problems. Convolutional Neural Networks (CNNs) have shown effectiveness in several dental imaging applications, like segmentation and categorization of anomalies in dental X-ray images, according to an increasing amount of research. The Tufts Dental Database, a large dataset of 1000 panoramic dental radiographs painstakingly categorized by experts, is one important addition to this order. Traditional methods for analyzing dental images frequently depend on machine learning techniques that require complex feature engineering and preprocessing. Although these techniques have succeeded, they face issues with efficiency, performance, and scalability when applied to larger datasets. Recent advancements in deep learning, including attention mechanisms and transformers, have revolutionized this field, enabling neural networks to learn directly from raw data and overcome these challenges. This paper aims to improve upon existing methods by developing an automated tooth segmentation system based on the U-Net++ architecture, modified with attention mechanisms to enhance segmentation accuracy in panoramic radiographs. Unlike traditional approaches, the proposed model directly leverages deep learning to improve segmentation precision, thereby addressing the limitations of prior methods. By using advanced neural network designs and preprocessing techniques, the model demonstrates superior segmentation performance, paving the way for more accurate and efficient dental treatment models and diagnostics.

Paper Organization: The rest of the paper is organized as follows. In Section II, we present a detailed review of related work in the field of automated tooth segmentation and the various deep learning models used. Section III describes the methodology, including the architecture of the proposed U-Net++ model and the integration of the

Efficient Channel Attention (ECA) mechanism. Section IV outlines the experiments conducted, including dataset preparation, training procedures, and performance metrics. In Section V, we discuss the results, compare the proposed method with existing techniques, and highlight its advantages. Finally, Section VI concludes the paper and suggests possible directions for future research.

II. RELATED WORK

Leea et al. [1] sought to determine the effectiveness of deep convolutional neural networks (CNNs) in differentiating dental cavities from periapical X-rays. In total, three thousand photographs were used. Regarding pretext processing, CNN is used and Inception V3 for transfer learning. They demonstrated a high diagnostic performance in molar, premolar, and combined configurations and identified that the premolar configuration performed the best in terms of the AUC. Hussain et al. [2] employed several state-of-the-art techniques on deep learning to segment brain tumors. Their proposed architecture predicts the outputs for corresponding labels and their algorithm works for the over-fitting. The work also employed the pre-processing steps such as N4ITK bias field correction. Stember et al. [3] compared the manual annotations with segmentation masks derived from data logged by eye-and-hand-recording equipment used for deep learning in medical imaging. They analyzed 19 radiologic screenshots and 356 T1-weighted contrast-enhanced meningioma pictures and proved that both masks are significantly different from each other with very high Dice similarity coefficient. The outcomes showed that although there are differences between the two types of masks, they do not significantly affect the CNNs performance, and further explained why eye tracking is so useful for deep learning segmentation.

Muresan et al. [4] described one new method, which is developed over deep learning and advanced image processing to understand the location of teeth and categorize the various dental abnormalities in panoramic X-ray images. A dataset including 14 distinct dental illnesses and after using annotated data to train their model for semantic segmentation, they segmented and improved the bounding boxes that corresponded to the tooth determinations using a variety of image processing operations. Within the discovered region of interest, each tooth occurrence was tagged and the issue impacting it was determined by a majority vote method based on a histogram. Panetta et al [5] employed a dataset consisting of one thousand panoramic dental radiography pictures. The researchers categorized these images into five categories: Peripheral features, radiodensity, appendicular load, effect on neighbouring tissues or organs and abnormality profile. This dataset's main goal is to improve tooth segmentation algorithms. Notably, in their studies, using convolutional neural network produced the best accuracy.

Sultan et al. [6] utilized panoramic images of radiographs for performing teeth segmentation with the help of a noise-reduced encoder-based residual U-Net model. In this paper, the authors pre-processed the dataset that contains 1000 images and then used their proposed residual U-net CNN

model which tries to predict the teeth defects using the denoised encoder block of the model. Ghahremani et al. [7] separated cavities and dental restorations in X-ray pictures by using convolutional neural networks. Using the Universal Teeth Numbering (UTN) system, the procedure entails determining the quantity and placement of each tooth. They achieved great accuracy and recall by addressing problems with teeth ordering and prediction in X-ray images. Next, each tooth's picture was cropped so that the cavity and repair are precisely segmented. Through the identification of abnormalities that individuals might not detect, their strategy lessens responsibility and lowers prejudice in treatment assessment.

Merter et al. [8] proposed deep learning models for the segmentation of teeth and maxillomandibular region. Their work utilizes medical image segmentation techniques such as Segmenter, Vision Transformer (ViT), and U-Net, which outperformed all the other benchmark models. EhaRenwi et al. [9] used a U-net model for the task of segmenting images of the teeth from the X-ray. On the same network, U-Net was also employed to perform data pre-processing by normalizing gamma plus applying histogram equalization. On the same note, U-Net was better off than the other deep learning structures.

Csaba et al. [10] explored the use of convolutional neural networks especially U-Net, for dental radiology's medical image segmentation. They concentrated on the difficult task of segmenting dental restorations, including fillings, crowns and root canal treatments, on panoramic radiographs because of their varying shapes and sizes. 1781 panoramic radiographs were used that specialists annotated for various dental restorations tasks. For training, the dataset was divided into multiple tiles, with the number of tiles ranging from 2 to 20. An encoder-decoder structure with skip links made up the model. With increase in number of tiles utilized, the F1-score increased, demonstrating major improvements in segmentation accuracy, particularly for root canal fills. SenbaoHou et al. [11] discussed the U-Net concept and the use of the Squeeze-Excitation Module in the integration of the encoder and the decoder. They proved that in order to overcome semantic gap, one needs to employ thick skip connections. Besides, they got features regarding the shapes of teeth with an aid of Multi-scale Aggregation Attention Block (MAB). At the bottleneck layer, they established a Dilated Hybrid Self-Attentive Block (DHAB) that is specifically capable of disregarding irrelevant background details.

Çelik et al. [12] used ten different models for identifying periapical lesions investigation. Furthermore, the effectiveness of these models was assessed on other datasets. According to the research, periapical lesions on periapical radiographs (PRs) may be reliably predicted by the deep learning models RetinaNet and adaptive training. Silva et al. [13] discussed the challenges that dentists faced when evaluating X-rays and proposed computational tools to help with automated tooth identification and categorization. They concentrated on using deep learning techniques, specifically four network frameworks which are Mask R-CNN, ResNeSt, PANet, and HTC to divide and recognize teeth in panoramic X-rays. PANet performed the best in segmentation and numbering.

Chahal et al. [14] integrated two-pathway and cascade structures for the analysis to learn spatial, depth-wise features and dependencies of neighboring labels for more accurate segmentation for implementation of brain segmentation. There are two types of cascade frameworks used: MFC Cascade and Input Cascade. The MFC Cascade performs worse than the Input Cascade. The proposed methodology involves slicing 3D MRI scans into 2D slices for processing, addressing class imbalance issues due to fewer tumor slices compared to non-tumor slices.

Ali et al. [15] aimed to provide a fuzzy clustering algorithm for the segmentation of dental X-ray images, based on indeterminate orthogonal matrices. Their work discusses the difficulties caused by shared cluster borders and emphasizes the significance of accurate segmentation in medical imaging for successful diagnosis. The proposed algorithm uses orthogonal principles to create pixel clusters and transforms image data into indeterminate representation sets in order to improve segmentation accuracy. The outcomes showed that the novel approach outperformed the most advanced techniques for quality clustering. Several other efficient algorithms have been proposed to handle different healthcare problems can be studied by referring to [16-19].

III. METHODOLOGY

A. Dataset Description

The work uses Tufts Dental Dataset, a dataset consisting of real-time dental X-ray images provided by Tufts University to comprehensively analyze the performance of the model.

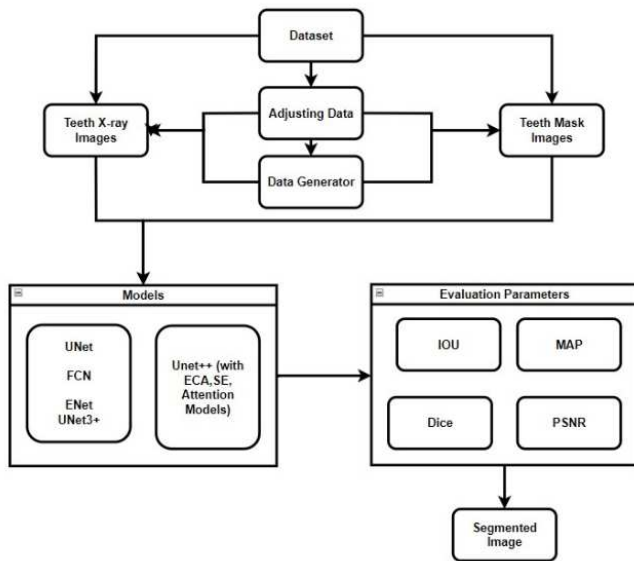


Fig. 1. Overview of the proposed study

It includes 1,000 panoramic dental radiographs, each professionally annotated with details on teeth and anomalies. The dataset folder contains X-ray images, teeth masks, maxillomandibular masks, and abnormality masks. Five aspects were used to classify these images: anatomical location, peripheral features, radiation density, and impacts on the surrounding structure. It has images of different patients' teeth and dental architecture, which enables researchers to create and evaluate algorithms for dental problem diagnosis. The collection includes pictures captured

at various resolutions, as well as a range of dental diseases. Additionally, the data set is used for additional dental diagnostic applications such as bone structure analysis, cavity identification, and tooth segmentation.

B. Proposed Methodology

The methodology proposed in the paper develops a novel combination of U-Net++ model and Efficient Channel Attention mechanism for enhanced teeth segmentation. For the selection of U-Net++ as the backbone of the proposed segmentation model, the study implemented six other state-of-art segmentation models to identify the best performing baseline model on the underlying dataset.

U-Net++ is derived from the simple U-Net, with additional convolutional layers and dense skip connections to improve the feature extraction of different details and merging them respectively. The architecture features alternating contracting and expansive pathways. The contracting path applies convolution and pooling operations for the prediction of contextual information, while the expansive path includes up-sampling and concatenation to predict the pixel location. Interlayer jump connections that define how one layer connects to the subsequent layer help in maintaining location information to enable accurate segmentation.

As presented in Figure 2, the model includes five blocks in the encoding phase and decoding phase. In encoding phase each block is a combination of convolutional layers followed by batch normalization and max pooling layers. Max-pooling layers are used for downsampling and dropout layers for regularization. The extraction of the bottleneck layer is done at a very high level of abstraction. In the decoding or the expansive path the upsample of the feature map by way of transposed convolution to the output resolution connect skip connections to concatenate features from the encoder or contracting path before convoluting the data.

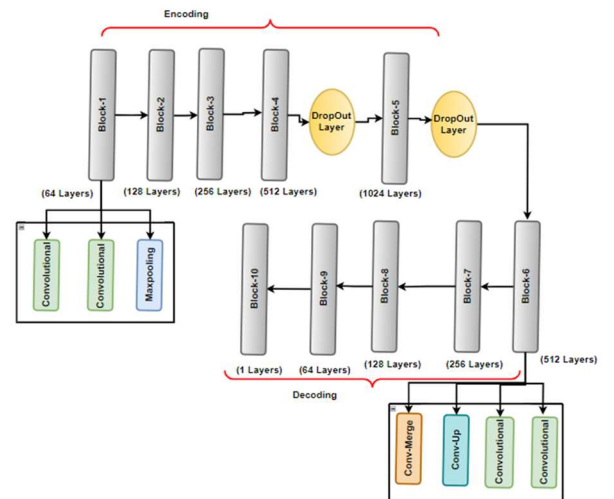


Fig. 2. U-Net++ Architecture

C. Efficient Channel Attention Network:

The baseline U-Net++ model is leveraged by integrating the Efficient channel attention network layer into the UNet++ model at its most complex level, named the bottleneck as

presented in Figure 3. Efficient Channel Attention Network (ECA-Net) is a lightweight and effective attention mechanism designed to improve feature representation by focusing on important channel-wise information. For downsampling the feature maps to have a single value per channel, it applies global average pooling layer followed by a 1D convolution layer to capture the interaction at local cross-channel level. Last but not the least, it computes the attention weights by employing sigmoid activation followed by the use of these weights to scale the input feature. This attention mechanism helps to achieve better feature representation because it allows to focus on the most informative channel. It is necessary to normalize and resize the pictures and masks during the training process.

Due to blending with the ECA-Net layer, the model can capture small details and get a very large boost in segmentation performance without suffering a very high computational load. UNet++ enhances the feature maps through the channel attention block applied channel-wise after the ECA layer post-bottleneck. We can assist segmentation attempts through the usage of the weighted feature maps as it focuses on the relevant data. The resulting maps are then up-sampled before a further segmentation step. This makes the ECA-Net integrated UNet++ a very effective and potent architecture for medical picture segmentation and associated tasks to leverage on the improved feature representation and the effective channel wise attention for enhanced performance.

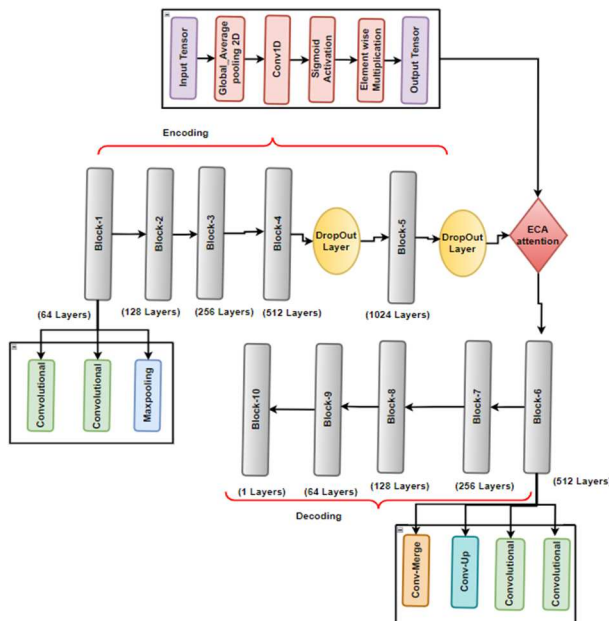


Fig. 3. The proposed UNet++ model integrated with Efficient Channel Attention Network (ECA-Net) for teeth segmentation.

D. PILOT STUDY

For the purpose of this study, the work implements different state-of-art segmentation models for baseline model selection and comprehensive comparisons to prove the robustness of the proposed model

U-shaped neural network (U-Net):

The U-Net architecture features alternating contracting and expansive pathways. The contracting consists convolution

layers followed by pooling layers that aids in predicting the contextual information. On the other hand, the expansive path includes up-sampling layers and concatenation layers to predict the pixel location. Interlayer jump connections connecting the layers to the subsequent layers maintain location information to enable accurate segmentation. U-Net effectively achieves one-to-one mapping of objects in images due to its capability to learn from sparse training and thereby segment images for a number of applications.

Fully Convolutional Networks (FCNs):

Fully Convolutional Networks (FCNs) are designed for semantic segmentation, performing pixel-level segmentation through an end-to-end approach with a backbone architecture. For the input images, they employ the convolutional layers to perform the density prediction of the images. FCNs perform segmentation tasks by preserving aspects of spatial information and adapting well to input image size. This makes them applicable to many images where need for segmentation of pictures is of the essence particularly when the spatial topography is critical.

Efficient Neural Network (ENet):

Efficient Neural Network (ENet) is developed especially to perform real-time semantic segmentation in scenarios with limited hardware resources. It uses bottleneck layers, asymmetric convolutions, and deep separable convolutions for achieving a high degree of segmentation and reduced computational costs. Specifically, ENet also aggregates spatial and channel-wise features for accurate pixel-level context description in a small window footprint, so that efficient pixel-wise predictions can be made.

U-Net3+

U-Net 3+ adds multi-resolution merging, attention mechanisms to the original U-Net architecture. It includes convolutional dense blocks and feature pyramid attention modules for enhancement of features agreement and for acquisition of multi-scale collective context information. Even more, the usage of a global context module increases the awareness of the scene as a whole and results in increased accuracy and entity segmentation repeatability.

SwiftNet:

SwiftNet is a convolutional neural network for real-time semantic segmentation where the resource efficient operations are made suitable for edge devices. It utilizes the depth-wise separable convolution type and efficient spatial pyramid pooling to obtain high visibility with low computational requirements. To overcome these challenges SwiftNet integrates contexts at multi-scales through a feature fusion technique which minimizes energy consumption and the response time of the machines involved. This makes it ideal for a variety of applications in edge computing

IV. EXPERIMENTATION AND RESULTS

The input images are initially normalized and thresholded to obtain binary masks. An additional subclass to the DataGenerator class is created to handle the images and

masks in batches. All the models implemented are trained using the same parameter settings to maintain consistency in model learning environments. Binary cross entropy loss and Adam optimizer are used by the models to learn the dataset features efficiently with a learning rate of 0.0001. The models are trained for 100 epochs and are compounded with early stopping so as to reduce both training and validation loss and to avoid over-fitting of the models.

The models evaluate their segmentation performance based on the set of standard metrics namely, Accuracy, Intersection over Union, Dice Coefficient, Peak Signal to Noise ratio, F-score, and Mean average Precision (mAP) performance metrics. Higher values of accuracy, IoU, dice coefficient, peak signal to noise ratio (PSNR), F-score and mAP indicate good results. Dice Coefficient is used in image segmentation measures similarity between predicted and ground truth masks, PSNR quantifies the quality of a compressed or reconstructed image by measuring the ratio between the maximum signal power and the power of noise, mAP measures a model's performance in object detection by averaging the average precision across classes and IoU measures the overlap between the predicted and ground truth masks in object detection and segmentation.

TABEL I. TRAINING, VALIDATION AND TESTING LOSS METRICS OBTAINED BY DIFFERENT MODELS ON THE TOOTH SEGMENTATION DATASET.

Models	Train		Validation		Test	
	Loss	Acc	Loss	Acc	Loss	Acc
U-Net	0.223	0.906	0.239	0.901	0.234	0.902
FCN	0.174	0.911	0.171	0.913	0.176	0.908
ENet	0.191	0.915	0.317	0.895	0.293	0.899
UNet++	0.063	0.975	0.074	0.970	0.072	0.970
U-Net3+	0.183	0.913	0.198	0.906	0.192	0.908
Swift Net	0.592	0.757	0.609	0.683	0.615	0.680
UNet++ ECA Net (Proposed)	0.046	0.981	0.053	0.979	0.053	0.978

Table I presents the loss and the accuracy values obtained by the different models for training, validation and testing phases over epochs. It can be seen that all the state-of-art models performed decently on the dataset but the SwiftNet model. It obtained the worst results with accuracies between 68% to 75%. It is clearly evident that amongst the baseline models, UNet++ achieved the best learning on the dataset, thereby, making it a strong candidate for the backbone of the proposed model. The proposed model that leverages the UNet++ model with ECA-Net attention mechanism obtains the highest accuracy and the least loss values for all the three data sets.

Figure 4 depicts the loss curve obtained for the proposed UNet++ model with ECA-Net attention mechanism. The graph demonstrates a good learning fit obtained by the model. The training loss and the validation loss reduced drastically during the initial 10 epochs with a gradual decrease between 10 to 20 epochs. The curve starts saturating after 20th epoch depicting a good model fit on the dataset. It is evident that ECA-Net attention model with U-

Net++ can perform high-quality segmentation for the input modality.

Next, the comparisons amongst different models in Table II proves that the proposed UNet++ model enhanced with Efficient channel attention performs the best over-all based on PSNR, Dice coefficient, mAP and accuracy metrics. UNet++ with ECA Net achieves a Dice coefficient of 0.907 and a PSNR of 16.77, suggesting strong segmentation accuracy and high-quality image retention. Also, UNet++'s selection as a baseline model is supported by a remarkable map=92% and Dice coefficient = 841.6%.

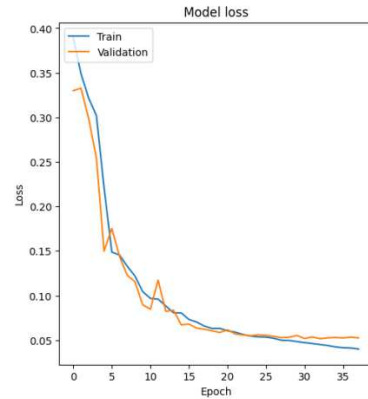


Fig. 4. Loss curve obtained by the proposed UNet++ with ECA-Net attention mechanism for tooth segmentation.

UNet++ with ECA-Net mechanism's higher metrics underscore its superior ability to accurately segment images and retain image quality. The overall results demonstrate that while attention mechanisms significantly enhance the UNet++ architecture, the ECA attention mechanism is the preferred choice for automated tooth identification and segmentation in X-ray images with the highest accuracy of 97%.

TABEL II. PERFORMANCE COMPARISONS OF DIFFERENT MODELS AGAINST THE PROPOSED MODEL ON THE TOOTH SEGMENTATION DATASET.

Models	Loss	Dice	PSNR	mAP	Acc
U-Net	0.234	0.533	10.10	0.513	0.902
FCN	0.171	0.669	10.39	0.608	0.908
ENet	0.317	0.393	9.986	0.478	0.899
UNet++	0.054	0.816	18.04	0.920	0.970
U-Net3+	0.198	0.557	10.04	0.573	0.908
Swift Net	0.609	0.157	4.96	0.118	0.680
UNet++ with ECA Net (Proposed)	0.053	0.907	16.77	0.964	0.978

These metric values suggest that, the proposed model has enhanced generalization and segmentation ability due to the integration of ECA-Net mechanism. On the other hand, with the lowest Dice coefficient and PSNR scores, SwiftNet and the Efficient Neural Network appear to be unsuitable for

tooth segmentation while U-Net, Fully convolutional network, UNet3+ obtained very average performance with their Dice coefficient ranging between 50% to 60%.

Fig. 5 presents the graphical results of the segmentation mask generated by the proposed ECA-Net integrated UNet++ approach on the sample images from tooth segmentation dataset. It can be seen clearly that the segmentation masks generated by the proposed model is very much similar to the ground truth masks. Infact, sharper segmentation results are obtained that are able to distinguish between different teeth.

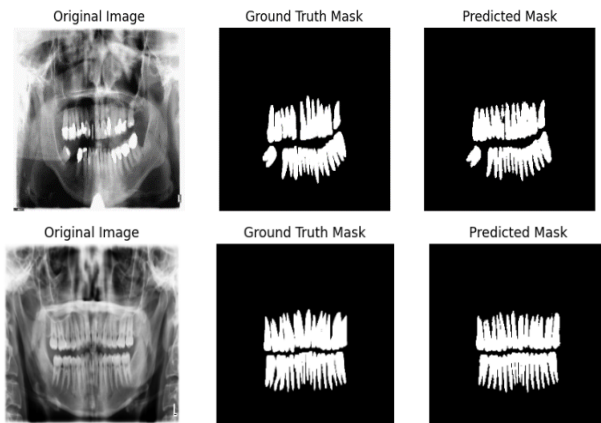


Fig. 5. Segmentation of Dental Xray using the proposed segmentation model

V. CONCLUSION AND FUTURE SCOPE

The work presented in this paper indicated that the baseline U-Net++ model consistently outperformed other models namely, U-net, FCN, ENet, U-net3+ and SwiftNet for dental segmentation by having low values of loss, high IoU, Dice coefficient values and high PSNR and mAP values. Further integrating U-Net with Efficient Channel Attention Network improved the model's performance, with the IoU of 0.830 and the Dice coefficient of 0.907. This implies enhanced segmentation skills and abilities in the respective marketing organizations. These conclusions all substantiate the generalization that the integration of enhanced attention mechanism improved segmentation and offers a solid and highly efficient solution for demanding segmentation. More research studies can be done to innovate and combined with other modern strategies like transformers, multi-head attention mechanisms etc. to enhance the efficiency of the segmentation in the future to enhance the diagnostic capacities of dentistry and contribute to creating better treatment strategies.

REFERENCES

- [1] Lee, J.H., Kim, D.H., Jeong, S.N. and Choi, S.H., 2018. Detection and diagnosis of dental caries using a deep learning-based convolutional neural network algorithm. *Journal of dentistry*, 77, pp.106-111.
- [2] Sajid, S., Hussain, S. and Sarwar, A., 2019. Brain tumor detection and segmentation in MR images using deep learning. *Arabian Journal for Science and Engineering*, 44, pp.9249-9261.
- [3] Stemmer, J.N., Celik, H., Krupinski, E., Chang, P.D., Mutasa, S., Wood, B.J., Lignelli, A., Moonis, G., Schwartz, L.H., Jambawalikar, S. and Bagci, U., 2019. Eye tracking for deep learning segmentation using convolutional neural networks. *Journal of digital imaging*, 32, pp.597-604.
- [4] Muresan, M.P., Barbura, A.R. and Nedeveschi, S., 2020, September. Teeth detection and dental problem classification in panoramic X-ray images using deep learning and image processing techniques. In *2020 IEEE 16th International Conference on Intelligent Computer Communication and Processing (ICCCP)* (pp. 457-463). IEEE.
- [5] Panetta, K., Rajendran, R., Ramesh, A., Rao, S.P. and Agaian, S., 2021. Tufts dental database: a multimodal panoramic x-ray dataset for benchmarking diagnostic systems. *IEEE journal of biomedical and health informatics*, 26(4), pp.1650-1659.
- [6] Almalki, S.A., Alsubai, S., Alqahtani, A. and Alenazi, A.A., 2023. Denoised encoder-based residual U-net for precise teeth image segmentation and damage prediction on panoramic radiographs. *Journal of Dentistry*, 137, p.104651..
- [7] Ghahremani, T., Hoseyni, M., Ahmadi, M.J., Mehrabi, P. and Nikoofard, A., 2023, December. Advanced Deep Learning-Based Approach for Tooth Detection, and Dental Cavity and Restoration Segmentation in X-Ray Images. In *2023 11th RSI International Conference on Robotics and Mechatronics (ICRoM)* (pp. 701-707). IEEE.
- [8] Karacan, M.H. and Yücebas, S.C., 2022, June. A deep learning model with attention mechanism for dental image segmentation. In *2022 International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)* (pp. 1-4). IEEE.
- [9] Fariza, A., Arifin, A.Z. and Astuti, E.R., 2020, October. Automatic Tooth and Background Segmentation in Dental X-ray Using U-Net Convolution Network. In *2020 6th International Conference on Science in Information Technology (ICSITech)* (pp. 144-149). IEEE.
- [10] Rohrer, C., Krois, J., Patel, J., Meyer-Lueckel, H., Rodrigues, J.A. and Schwendicke, F., 2022. Segmentation of dental restorations on panoramic radiographs using deep learning. *Diagnostics*, 12(6), p.1316.
- [11] Hou, S., Zhou, T., Liu, Y., Dang, P., Lu, H. and Shi, H., 2023. Teeth U-Net: A segmentation model of dental panoramic X-ray images for context semantics and contrast enhancement. *Computers in Biology and Medicine*, 152, p.106296.
- [12] Çelik, B., Savaştaer, E.F., Kaya, H.I. and Çelik, M.E., 2023. The role of deep learning for periapical lesion detection on panoramic radiographs. *Dentomaxillofacial Radiology*, 52(8), p.20230118.
- [13] Silva, B., Pinheiro, L., Oliveira, L. and Pithon, M., 2020, November. A study on tooth segmentation and numbering using end-to-end deep neural networks. In *2020 33rd SIBGRAPI conference on graphics, patterns and images (SIBGRAPI)* (pp. 164-171). IEEE.
- [14] Chahal, E.S., Haritosh, A., Gupta, A., Gupta, K. and Sinha, A., 2019, October. Deep learning model for brain tumor segmentation & analysis. In *2019 3rd International conference on recent developments in control, automation & power engineering (RDCAPE)* (pp. 378-383). IEEE.
- [15] Ali, M., Khan, M. and Tung, N.T., 2018. Segmentation of dental X-ray images in medical imaging using neutrosophic orthogonal matrices. *Expert Systems with Applications*, 91, pp.434-441.
- [16] Resmi, S., Singh, R.P. and Palaniappan, K., 2024. Automated Cervical Cytology Image Cell Segmentation Using Enhanced MultiResUNet with DCT and Spectral Domain Attention Mechanisms. *IEEE Access*.
- [17] Resmi, S., Singh, R.P. and Palaniappan, K., 2024. Automatic Skull Shape Completion of Defective Skulls Using Transformers for Cranial Implant Design. *Procedia Computer Science*, 235, pp.3305-3314.
- [18] Garrepalli, V., Gupta, V., Abhishek, S. and Anjali, T., 2024, May. Empowering oral healthcare: Cnn-assisted diagnosis for improved patient outcomes. In *2024 Second International Conference on Data Science and Information System (ICDSIS)* (pp. 1-5). IEEE.
- [19] Das, P., Thimmaraju, M.K., Ahalya, N., Anandaram, H. and Kumar, A., 2022, October. Design and Comparison of Transfer Learning for Dental Caries Detection. In *2022 International Conference on Edge Computing and Applications (ICECAA)* (pp. 1311-1316). IEEE.